**COMP 2711 – Mock Midterm 10**

# Solutions and marking notes

Practice material – not an official examination

## Problem 1 [10 points]

(a) **[3 points]** (i) Tautology: $(\neg p \vee q) \vee (\neg q \vee p)$ contains complementary pairs. (ii) Contradiction, using the XOR/negated-biconditional equivalence and negation law:

$$(p \leftrightarrow q) \wedge (p \oplus q) \equiv (p \leftrightarrow q) \wedge \neg(p \leftrightarrow q) \equiv F.$$

(iii) Contingency: it simplifies to $\neg p \wedge \neg q$, true when both are F and false when both are T.

*Marking:* 1 each correct class with justification. A correct semantic justification also earns the classification mark.

(b) **[2 points]** The liar sentence is not a proposition: assigning either truth value makes its own assertion inconsistent. The second claim is also false: $q$ only if $p$ means $q \to p$, the converse. $p = F, q = T$ satisfies the given implication but refutes that converse.

*Marking:* 1 paradox classification/reason; 1 only-if direction and counterassignment.

(c) **[5 points]** (i) Write $u = a/b$, $v = c/d$ with integer numerators and $b, d \neq 0$. Since $v \neq 0$, $c \neq 0$. Thus $u/v = ad/(bc)$ is a ratio of integers with nonzero denominator. (ii) Write $n = 2k + 1$. Then $n(n + 2) = (2k + 1)(2k + 3) = 2(2k^2 + 4k + 1) + 1$, with integer witness $2k^2 + 4k + 1$, so the product is odd.

*Marking:* (i) 1 representations, 1 quotient, 1 c/nonzero denominator checks; (ii) 1 expansion, 1 odd integer witness.

## Problem 2 [13 points]

(a) **[4 points]** The grouping is $(\forall x P(x)) \vee Q(x)$. The $x$ in $P$ is bound and the $x$ in $Q$ is free, so the expression is open and not yet a proposition. The translation is $\exists y \in \mathbb{R}\, \forall x \in \mathbb{R}(x \leq y)$, false since $x = y + 1$ defeats any candidate $y$.

*Marking:* 2 grouping and free/open status; 1 translation; 1 candidate-dependent counterexample.

(b) **[4 points]**

$$\forall x(\neg P(x) \vee A) \equiv (\forall x \neg P(x)) \vee A \equiv \neg \exists x P(x) \vee A \equiv (\exists x P(x)) \to A.$$

The steps use implication, null quantification and quantifier negation; moving the antecedent quantifier reverses its type. On the empty domain take $A = F$: the left universal is vacuously T, whereas the right is $F \wedge T = F$.

*Marking:* 2 valid equivalence derivation/polarity; 1 empty-domain choice; 1 both side values.

(c) **[5 points]** Choose fresh $a$ satisfying $R(a) \wedge S(a)$ by existential instantiation. Simplify to $R(a), S(a)$. Instantiate $R(a) \to \neg Q(a)$ and use modus ponens to get $\neg Q(a)$. Instantiate $P(a) \to Q(a)$ and use modus tollens to get $\neg P(a)$. Conjoin $S(a) \wedge \neg P(a)$ and existentially generalize.
*Marking:* 1 fresh witness/simplification; 1 first UI and MP; 1 second UI; 1 modus tollens; 1 conjunction/EG.

**Problem 3 [12 points]**

(a) **[2 points]** $A \setminus B = \{0\}$ and $U \setminus (A \cup B) = \{1, 5\}$. Since $A \cap B = \{2, 4\}$, the union size is $3 + 3 - 2 = 4$, agreeing with the roster $\{0, 2, 3, 4\}$.
*Marking:* 1 both difference/complement rosters; 1 union size with intersection identity.

(b) **[3 points]**

$$h(n) = \begin{cases} n/2, & n \text{ even}, \\ -(n+1)/2, & n \text{ odd}. \end{cases}$$

Each branch is injective, and the branch ranges are disjoint, so $h$ is injective. Every $z \geq 0$ has preimage $2z$, and every $z < 0$ has preimage $-2z - 1 \in \mathbb{N}$, so it is onto.
*Marking:* 1 formula and integer outputs; 1 injection including disjoint ranges; 1 preimages for all integers.

(c) **[7 points]**

(i) **F**. No finite natural-number list contains every natural number.

(ii) **T**. The map $n \mapsto 2n$ is a bijection from $\mathbb{N}$.

(iii) **T**. Encode by six increasing positions and enumerate finite layers; varying the last position gives infinitely many sequences.

(iv) **T**. Filter an enumeration of rationals; $1/(n+2)$ gives infinitely many members.

(v) **T**. A proper subset omits at least one distinct element of the finite set.

(vi) **F**. Such a listing would make it countable.

(vii) **F**. Programs are finite strings over a finite alphabet and can be ordered by length.

*Marking:* Award 1 point per correct T/F choice (7 total); reasons are included here for checking.

**Problem 4 [16 points]**

(a) **[2 points]** The square-sum identity gives $\sum_{k=0}^{6} \binom{6}{k}^2 = \binom{12}{6} = 924$.
*Marking:* 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(b) **[2 points]** The triple forms one of six circular units, whose orders number 5!. Its internal orders number 3!, giving $5!3! = 720$.
*Marking:* 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(c) **[2 points]** The unrestricted count is $6!/(2!2!2!) = 90$. Compressing AA yields $5!/(2!2!) = 30$ forbidden strings. Thus $90 - 30 = 60$.
*Marking:* 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(d) **[2 points]** Subtract committees containing the designated pair: $\binom{10}{6} - \binom{8}{4} = 140$.
*Marking:* 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(e) **[2 points]** If the A-count is $2t$, for $t = 0, 1, 2, 3$, the remaining B,C counts have $7 - 2t$ possibilities. Sum: $7 + 5 + 3 + 1 = 16$.
*Marking:* 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(f) **[2 points]** Choose four of each term to obtain exponent zero, giving $\binom{8}{4}(-2)^4 = 1120$.
*Marking:* 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(g) **[2 points]** After fixing image 2, the first input can use only 3 or 4. The two remaining images are assigned in 2! ways, giving $2 \cdot 2! = 4$.
*Marking:* 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(h) **[2 points]** Total paths are $\binom{7}{3}$. A forbidden RR prefix leaves two right and three up steps, giving $\binom{5}{3}$. Subtract to get $35 - 10 = 25$.
*Marking:* 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

**Problem 5 [10 points]**

**[10 points]** In a $2n$-element ground set, complementation is an inverse-to-itself bijection from subsets of size below $n$ to subsets of size above $n$, because complementary sizes add to $2n$. Exactly $\binom{2n}{n}$ subsets have size $n$; they map within the middle-size class rather than between the small and large classes. Remove them from all $2^{2n}$ subsets. The remaining two classes have equal size, so the small class contains

$(2^{2n} - \binom{2n}{n})/2$ subsets. Partitioning that class by sizes $k = 0, \ldots, n-1$ gives the sum on the left.
*Marking:* 3 marks for the complement bijection and inverse; 2 for the middle-size class and its separation; 3 for subtracting and halving; 2 for the disjoint size-class sum.

**Problem 6 [9 points]**

**[9 points]** Let $A, B, C$ be the multiples of 2, 3 and 5. Apply IE to the three sets $A \cap B$, $A \cap C$, $B \cap C$. Their sizes are $30, 18, 12$. Every pairwise intersection and the triple intersection equals $A \cap B \cap C$, of size 6. Thus the count divisible by at least two is $30 + 18 + 12 - 3 \cdot 6 + 6 = 48$. Remove the six multiples divisible by all three, obtaining

$$30 + 18 + 12 - 3 \cdot 6 = 42.$$

Equivalently each exactly-two integer occurs once in the sum of pair counts, whereas a triple-divisible integer occurs three times and must be subtracted three times.
*Marking:* 2 marks for forming the three pair-intersection sets; 2 for their sizes; 2 for the IE union with common overlap; 2 for removing all-three integers exactly; 1 for 42.

**Problem 7 [7 points]**

**[7 points]** The one-count boxes are 0 through 8. Boxes 0 and 8 contain at most one distinct string each. If no box had three strings, the seven other boxes would hold at most two each, so at most $1 + 7 \cdot 2 + 1 = 16$ strings could be selected. Seventeen therefore force three equal-weight strings. To realize sixteen without a triple, take the unique weight-0 and weight-8 strings and two distinct strings of each weight 1 through 7. Each interior weight has at least two strings, for example ones in positions $1, \ldots, k$ or $2, \ldots, k+1$. Hence the construction is possible and the threshold is sharp.
*Marking:* 3 marks for recognizing endpoint capacities and the bound sixteen; 2 for the forcing conclusion; 2 for the explicit attainable sharp construction.

**Problem 8 [6 points]**

(a) **[1 points]** $P(A) = 1/8 + 1/8 = 1/4$ and $P(B) = 1/4 + 1/8 = 3/8$.
*Marking:* 1 mark for both marginal sums.

(b) **[1 points]** The intersection is $\{11\}$, with mass $1/8$.
*Marking:* 1 mark for $1/8$.

(c) **[1 points]** $P(A \cup B) = 1/4 + 3/8 - 1/8 = 1/2$.
*Marking:* 1 mark for the union mass.

(d) **[1 points]** By De Morgan, $A^c \cap B^c = (A \cup B)^c = \{00\}$, so its probability is $1/2$.
*Marking:* 1 mark for the complementary event mass.

(e) **[2 points]** $A \cup B$ is the disjoint union of $A \setminus B$, $B \setminus A$, and $A \cap B$. Adding their masses gives $P(A)+P(B)-P(A\cap B)$. Here $P(A)P(B) = 3/32 \neq 1/8 = P(A\cap B)$, so the events are dependent.
*Marking:* 1 mark for a valid disjoint decomposition and union identity; 1 for the independence comparison.

**Problem 9 [7 points]**

(a) **[3 points]** The conditioning event contains $\binom{4}{2} = 6$ strings. Three put a one first, giving $P(E \mid F) = 3/6 = 1/2$. The three adjacent placements $(1,2),(2,3),(3,4)$ give $P(G \mid F) = 3/6 = 1/2$.
*Marking:* 1 mark for the six-outcome restricted space; 1 for each correct favorable count/probability.

(b) **[4 points]** $E$ has eight strings, of which $\binom{3}{1} = 3$ contain exactly two ones. Hence $P(F \mid E) = 3/8 = P(F)$, establishing independence. The test $P(F \mid E) = P(F)$ is equivalent to joint factorization only when $P(E) > 0$; here $P(E) = 1/2$. If $P(E) = 0$, the conditional quotient is undefined.
*Marking:* 1 mark for restricted-space counting; 1 for equality with the marginal; 1 for the independence conclusion; 1 for the positive-denominator requirement.

**Problem 10 [10 points]**

(a) **[2 points]** $E(I_i) = p$ and $E(I_i^2) = p$, so $V(I_i) = p - p^2 = p(1-p)$.
*Marking:* 1 mark for the mean; 1 for the second-moment variance.

(b) **[2 points]** Linearity gives $E(S) = np$. Independent variance addition gives $V(S) = np(1-p)$. Only the latter addition requires the stated independence.
*Marking:* 1 mark for each result with the correct independence role.

(c) **[2 points]** $E(M) = np/n = p$ and $V(M) = np(1-p)/n^2 = p(1-p)/n$. The mean is preserved while the variance is divided by n.
*Marking:* 1 mark for each result with correct scaling.

(d) **[4 points]** Put $\mu = E(X)$. Expanding centered squares gives

$$V(X) = E(X^2 - 2\mu X + \mu^2) = E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - \mu^2.$$

With $\nu = E(Y)$, independence gives $E[(X-\mu)(Y-\nu)] = E(X-\mu)E(Y-\nu) = 0$.

Thus

$$\begin{aligned} V(X+Y) &= E\big([(X-\mu)+(Y-\nu)]^2\big) \\ &= V(X) + 2\cdot 0 + V(Y) = V(X) + V(Y). \end{aligned}$$

*Marking:* 2 marks for the centered-square expansion and variance identity; 2 for the vanishing centered cross-product and addition proof.